

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization

M. Ibrahim, F. Janabi-Sharifi, A. Mohammadi, U. Stege, and P. Angara

Toronto Metropolitan University, Toronto, ON, Canada; University of Victoria, Victoria, BC, Canada

Main contact: M. Ibrahim, ibrahimshameel89@gmail.com

I. POSTER TITLE

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization.

II. POSTER AUTHORS

M. Ibrahim, F. Janabi-Sharifi, and A. Mohammadi are with Toronto Metropolitan University, Toronto, ON, Canada. U. Stege and P. Angara are with the University of Victoria, Victoria, BC, Canada. The main contact is M. Ibrahim, ibrahimshameel89@gmail.com.

III. POSTER ABSTRACT

Trajectory optimization for UAV target tracking in cluttered environments must jointly handle obstacle avoidance, target visibility, and terminal-goal constraints. This poster presents a discrete Higher-Order Unconstrained Binary Optimization (HUBO) formulation for obstacle-avoidance and occlusion-aware UAV trajectory planning, together with a feasibility-aware evaluation protocol for soft-penalty quantum-optimization benchmarks. The work does not claim quantum speedup. Instead, it asks whether a higher-order binary formulation captures meaningful planning structure and how penalty-based solvers should be evaluated. Feasibility-aware benchmarking means that solver outputs are decoded as trajectories and checked for one-hot occupancy, start, motion, obstacle, and terminal constraints, rather than judged only by total penalized energy. On an 8×8 instance with 960 original binary variables, 87,684 monomials, and 3,172 cubic terms, we compare a classical A* planner, simulated annealing, D-Wave/Ocean neal after quadratization to a binary quadratic model, and Qiskit QAOA on a reduced quadratized instance. A* and simulated annealing obtain comparable best soft costs, but A* is far faster and always feasible. Neal is fast after quadratization but yields low decoded feasibility unless penalty weights are substantially increased. The poster highlights the need for constraint-aware validation when quantum-classical optimization methods are used for robotics-inspired planning benchmarks.

IV. POSTER RELEVANCE

The poster is relevant to QCE topics in hybrid quantum-classical computing, adiabatic quantum computation and quantum annealing, QAOA, variational and empirical algorithm analysis, quantum applications in optimization and decision making, and quantum systems software benchmarking. It connects a robotics-inspired planning problem to quantum-compatible optimization models while emphasizing practical validation: decoded feasibility, qubit/logical-variable growth after quadratization, solver-specific assumptions, and comparison with classical baselines.

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization

M. Ibrahim, F. Janabi-Sharifi, A. Mohammadi, U. Stege, and P. Angara

Toronto Metropolitan University, Toronto, ON, Canada; University of Victoria, Victoria, BC, Canada

Abstract—Trajectory optimization for UAV target tracking in cluttered environments must jointly address obstacle avoidance, visibility, and terminal-goal constraints. This extended abstract presents a discrete higher-order unconstrained binary optimization (HUBO) model for obstacle-avoidance and occlusion-aware UAV trajectory planning and a feasibility-aware benchmarking protocol for soft-penalty quantum-optimization instances. The purpose is not to claim quantum speedup. Rather, the contribution is a benchmark formulation, a classical baseline, and a validation method that distinguishes the penalized objective value from decoded-trajectory feasibility. The full 8×8 instance contains 960 binary trajectory variables, 87,684 monomials, and 3,172 cubic terms. A custom simulated annealing (SA) baseline evaluates the higher-order objective, while D-Wave/Ocean `neal` is treated correctly as a classical binary-quadratic-model (BQM) sampler and is used only after quadratization. Qiskit QAOA is evaluated only on a reduced quadratized instance as a correctness and compatibility check. A* and SA obtain comparable best soft costs (317 and 303), but A* runs in about 10 ms versus 47.8 s for SA. Neal is fast after quadratization but reaches only 22% true feasibility at the hardened weight setting. These results show why feasibility checks are necessary for penalty-based quantum-classical benchmarks.

Index Terms—HUBO, QUBO, quantum annealing, QAOA, A*, simulated annealing, UAV trajectory optimization, occlusion-aware tracking, feasibility-aware benchmarking.

I. INTRODUCTION AND PROBLEM SETTING

Visibility-aware UAV target tracking keeps a target in view while respecting environmental obstacles. Masnavi *et al.* formulated the corresponding occlusion-free target-tracking and obstacle-avoidance problem as a real-time multi-convex model predictive control (MPC) problem [1]. We use that work as the continuous-control reference and formulate a discrete soft-penalty benchmark suitable for quantum-optimization workflows.

Let V be the grid cells, $O \subset V$ the obstacle cells, $B \subset V$ a set of buffer cells near obstacles or visibility-critical regions, s the start cell, g_t the target cell at time t , and $N(u)$ the legal neighbor cells reachable from u in one step. A buffer cell is a cell where sustained residence is discouraged even if the cell is not strictly an obstacle. The binary variable $x_{t,v} \in \{0, 1\}$ equals one iff the UAV occupies cell v at time t . The

constrained planning problem is

$$\min_x C_{\text{vis}}(x) + C_{\text{prox}}(x) + C_{\text{term}}(x) \quad (1)$$

$$\text{s.t.} \quad \sum_{v \in V} x_{t,v} = 1 \quad \forall t, \quad (2)$$

$$x_{0,s} = 1, \quad x_{t,v} = 0 \quad \forall v \in O, \quad (3)$$

$$x_{t,u}x_{t+1,v} = 0 \quad \forall t, u \in V, v \notin N(u). \quad (4)$$

The continuous MPC dynamics and continuous visibility constraints are not reproduced here; the reader should consult [1]. The discrete benchmark replaces these constraints by penalties and then checks feasibility independently after decoding.

II. HUBO REFORMULATION

The penalty Hamiltonian is

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \quad (5)$$

Here $H_{\text{uniq}} = \lambda_u \sum_t (\sum_v x_{t,v} - 1)^2$ enforces one UAV cell per time step; $H_{\text{start}} = \lambda_s (1 - x_{0,s})$ fixes the start; $H_{\text{move}} = \lambda_m \sum_t \sum_{v \notin N(u)} x_{t,u}x_{t+1,v}$ penalizes invalid transitions; and $H_{\text{obs}} = \lambda_o \sum_t \sum_{v \in O} x_{t,v}$ forbids obstacle occupancy. We define $\text{LOS}(u, g_t)$ as the set of grid cells intersected by the Bresenham line segment from UAV cell u to target cell g_t . Occlusion-aware means that obstacle intersections with this line of sight affect the cost:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_{u \in V} x_{t,u} |\text{LOS}(u, g_t) \cap O|. \quad (6)$$

For a static target this term is linear in the UAV variables; for a moving target encoded with binary target occupancy it becomes a joint product of UAV and target occupancy variables. The sustained-proximity term is genuinely cubic:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{b \in B} \sum_{t=1}^{T-2} x_{t-1,b} x_{t,b} x_{t+1,b}. \quad (7)$$

Finally, H_{term} enforces or rewards terminal arrival at the target or terminal region. The full 8×8 instance has 960 original binary variables and 3,172 cubic terms; a one-ancilla-per-cubic quadratization would yield approximately 4,132 logical BQM variables before any hardware embedding overhead.

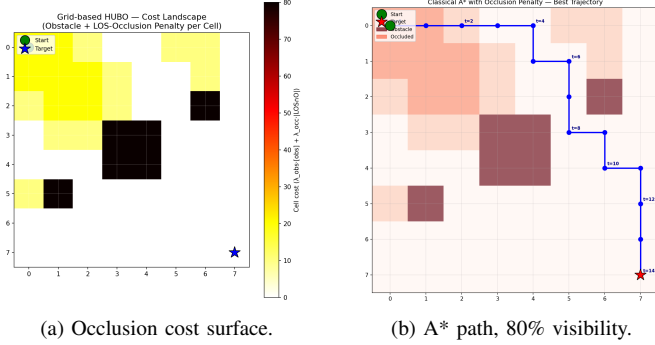


Fig. 1. Discrete occlusion-aware benchmark instance and a feasible A* trajectory.

TABLE I
FULL 8×8 INSTANCE, 960 ORIGINAL VARIABLES.

Metric	A*	SA	neal/BQM
Best soft cost	317	303	496
Feasibility rate	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	—	—
Runtime/sample	10 ms	47.8 s	0.26 s

III. FEASIBILITY-AWARE BENCHMARKING

This is a soft-penalty instance: a solver can obtain a low total penalized energy $H(x)$ while violating hard planning constraints. We therefore define *feasibility-aware benchmarking* as follows. Each sample is decoded into a trajectory and checked for one-hot occupancy, start, motion, obstacle, and terminal conditions. We separately report hard-penalty violations, best feasible soft cost, target-reaching, visibility, and runtime. In this paper, “raw energy” means the total penalized objective value in (5) before this decomposition.

The solvers are used in their appropriate regimes. A* is a classical graph-search baseline [4]. SA refers to simulated annealing [5] applied to the benchmark objective. D-Wave/Ocean *neal* is not a quantum solver and not a direct HUBO solver; it is a classical simulated-annealing BQM/QUBO sampler, so the higher-order model must be quadratized before it is sampled [7], [8]. Similarly, the Qiskit Optimization path used here applies QAOA to a quadratic representation on a tiny instance, not to the full HUBO [6], [9]. Direct k -local QAOA for higher-order terms is possible in principle but was not the implementation tested here.

IV. RESULTS

Table I reports the matched comparison on the full 8×8 instance using 50 matched stochastic samples, SA at 500 sweeps, and hardened weights $\lambda = 1000$. A* and SA reach comparable best soft costs (317 and 303), but A* solves the planning analogue in about 10 ms while SA requires 47.8 s. Neal is fast after quadratization, but only 22% of its samples are feasible and its best feasible soft cost is worse. A penalty sweep further shows that neal feasibility increases from 0% at $\lambda = 50$ to 60% at $\lambda = 2000$, demonstrating why single-weight reports can overstate solver quality.

On the reduced 2×2 demonstrator, the QAOA experiment uses eight logical binary variables after quadratization and is included as a correctness/compatibility check. A*, SA, neal, and Qiskit QAOA all reach the same optimum $E^* = -118.17$. QAOA takes 55 s versus 0.02 s for neal and 0.14 s for SA, so it is not presented as a scalable performance result.

V. DISCUSSION AND CONCLUSION

The A* baseline answers whether the binary formulation preserves sensible planning structure: it finds a feasible, target-reaching, visibility-aware trajectory in milliseconds. The value of the HUBO formulation is instead its compatibility with higher-order and quadratized quantum-optimization workflows and its extensibility to moving targets, where the line-of-sight interaction can become intrinsically higher-order. The key methodological conclusion is that penalty-model benchmarks should be evaluated by decoded feasibility, not by raw penalized energy alone.

Future work will verify and release the implementation, compare against the continuous MPC baseline of [1], test moving-target instances, and evaluate quadratized models on real quantum annealing hardware.

REFERENCES

- [1] H. Masnadi *et al.*, “Real-Time Multi-Convex MPC for Occlusion-Free Target Tracking With Quadrotors,” *IEEE Access*, vol. 10, pp. 113537–113550, 2022.
- [2] N. Innan *et al.*, “QUAV: Quantum-assisted path planning and optimization for UAV navigation with obstacle avoidance,” arXiv:2508.21361, 2025.
- [3] E. Osaba *et al.*, “Solving drone routing problems with quantum computing,” arXiv:2501.18432, 2025.
- [4] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Trans. Syst. Sci. Cybern.*, 1968.
- [5] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, “Optimization by simulated annealing,” *Science*, 1983.
- [6] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” arXiv:1411.4028, 2014.
- [7] D-Wave Systems, “dwave-neal: SimulatedAnnealingSampler documentation,” 2021.
- [8] D-Wave Systems, “dimod make_quadratic and BQM documentation,” Ocean SDK documentation, 2026.
- [9] Qiskit Community, “Qiskit: An open-source framework for quantum computing,” Zenodo, doi:10.5281/zenodo.2562111, 2019.

Feasibility-Aware Quantum-Classical Benchmarking

of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization

M. Ibrahim, F. Janabi-Sharifi, A. Mohammadi, U. Stege, and P. Angara

SCOPE

We introduce a quantum-compatible HUBO benchmark for occlusion-aware UAV trajectory optimization and a feasibility-aware methodology for evaluating such models. Classical baselines are deliberate: current quantum hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment chart a path toward future D-Wave / QAOA hardware studies.

1. The Problem

UAV target tracking in cluttered space must **simultaneously**:

- ▶ avoid obstacles, and
- ▶ keep **line-of-sight** to a target (occlusion-free).

Masnavi *et al.* solved this as continuous multi-convex MPC. We recast the **same problem class** as a discrete higher-order binary problem for quantum solvers.

2. HUBO Formulation

Binary $x[t, v] = 1$ iff UAV at cell v , time t . Seven cost terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}$$

Occlusion (Bresenham line-of-sight):

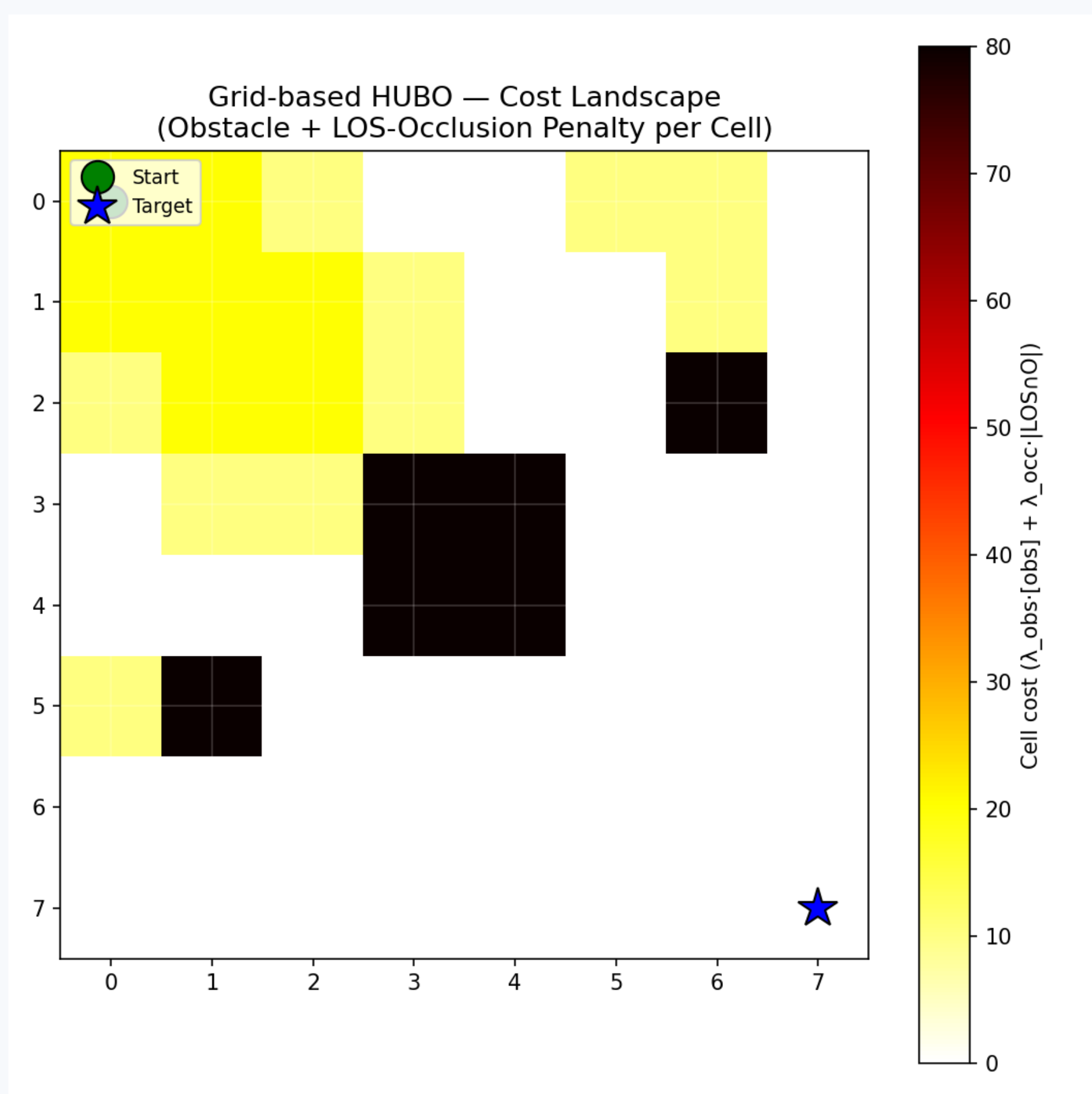
$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_t \sum_u x[t, u] | \text{LOS}(u, \text{tgt}) \cap O |$$

Genuinely cubic proximity (lingering in a buffer cell):

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b} \sum_t x[t-1, v_b] x[t, v_b] x[t+1, v_b]$$

Why HUBO not QUBO? Quadratizing the cubic term needs $\sim 3,172$ ancillas ($\sim 30\%$ more variables) with dense couplings that wreck sparsity.

Occlusion Cost Surface



Yellow = target occluded from that cell; dark = obstacle.

3. Feasibility-Aware Benchmarking

Penalized energy misleads — annealers lower energy by *violating* soft constraints. Four safeguards:

(i) Energy decomposition

hard-penalty vs. soft-cost, kept separate

(ii) Independent constraint check

decode + verify, not energy thresholds

(iii) Matched sampling budgets

equal reads/runs across solvers

(iv) Penalty-weight sweep

exposes the constraint-exploitation regime

Two-Tier Design

State-vector QAOA scales as 2^n — impossible at $n=960$.

TIER 1

Full 8×8
960 vars
 $A^* \cdot SA \cdot \text{neal}$

TIER 2

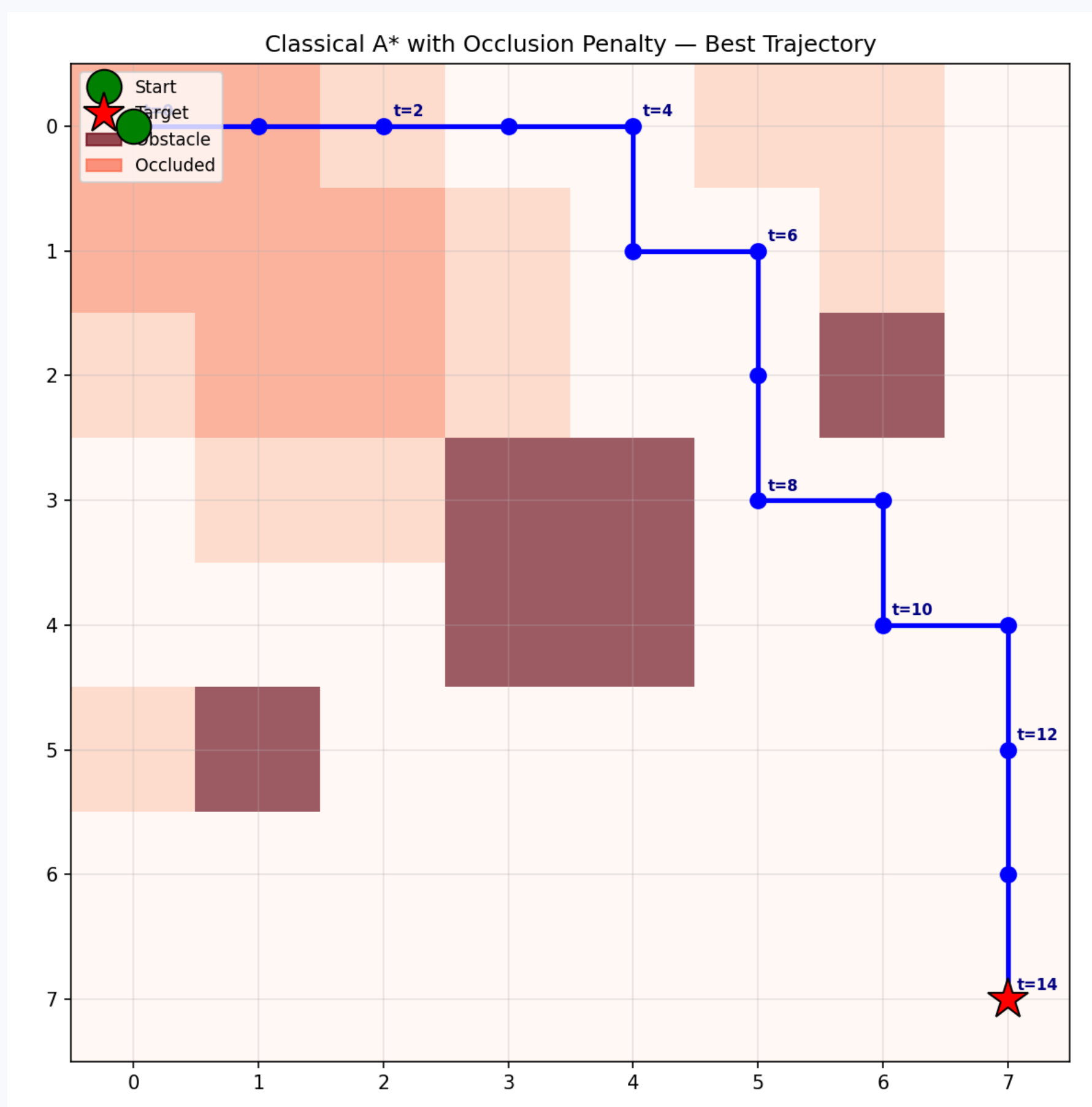
Tiny 2×2
8 vars
+ Qiskit QAOA

Classical A* Baseline

A* on the same grid with occlusion-weighted cell cost

$$1.0 + \lambda_{\text{occ}} | \text{LOS} \cap O | + \lambda_{\text{prox}} \mathbf{1}[v_b \in B]$$

grounds the HUBO against a conventional planner — answering whether the formulation captures **sensible planning** or is a benchmark artifact.

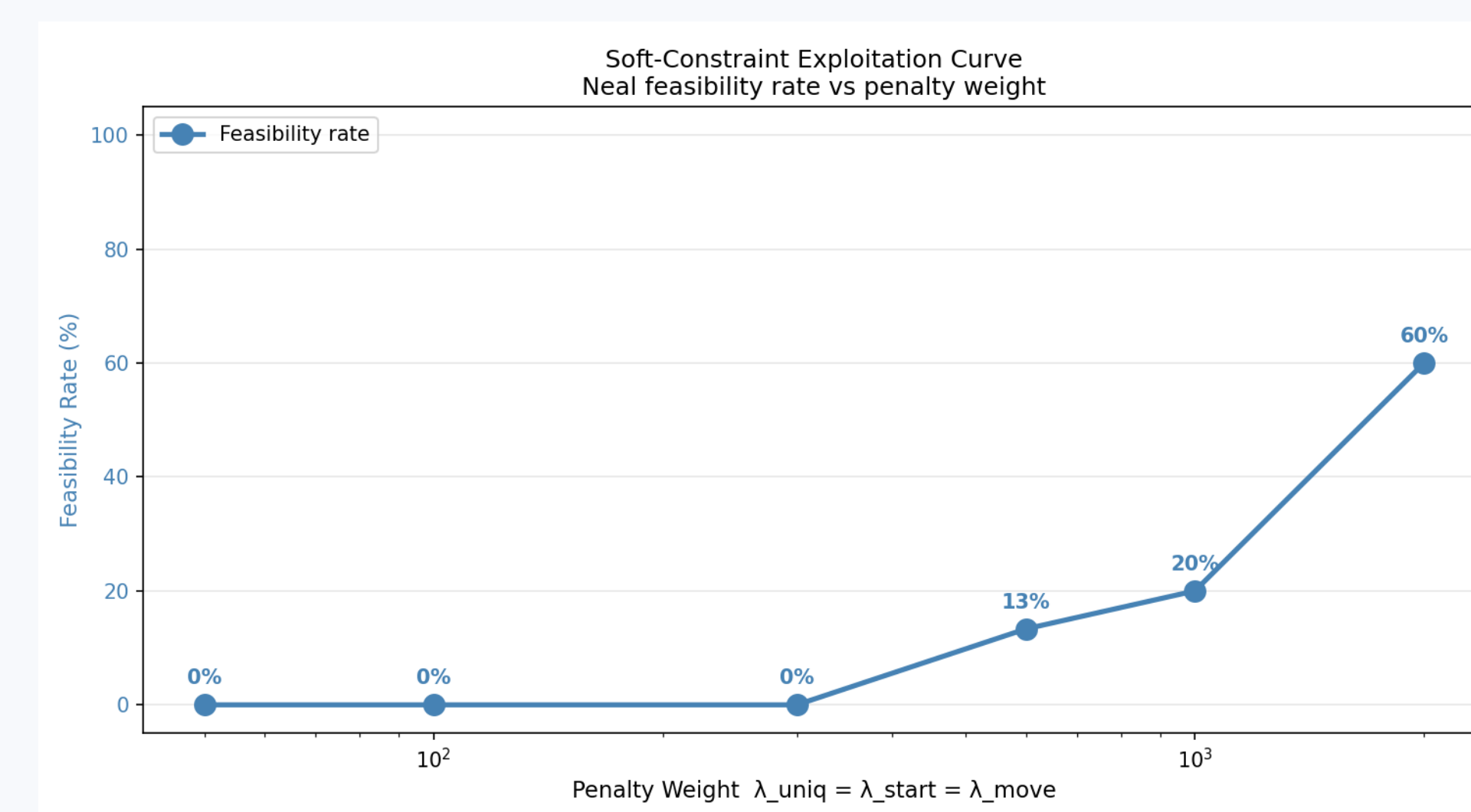


4. Results: 8×8 Instance

Metric	A*	SA	neal
Soft cost	317	303	496
Feasibility	100%	100%	22%
Reaches target	yes	14%	8%
Visibility	80%	—	—
Runtime/sample	10 ms	47.8 s	0.26 s

A* and SA reach comparable quality; A* is $\sim 4,700\times$ faster. neal is fast but only 22% feasible.

Headline: Constraint-Exploitation Curve



neal feasibility climbs **0% → 60%** as λ : 50 → 2000

General-purpose annealers need penalty weights an **order of magnitude** above the objective scale to respect one-hot HUBO structure. A single λ would misreport feasibility.

Tier 2 & Takeaways

On 8 variables, **all four solvers** reach the same optimum ($E^* = -118.17$); QAOA is a correctness check ($2,750\times$ slower, no scaling).

The HUBO's value is not raw speed here, but compatibility with quantum solvers and extensibility to a **moving target**, where occlusion becomes intrinsically cubic — beyond A*'s shortest-path reach.

Future: moving target · real D-Wave hardware · vs. continuous MPC.

CONTACT + DOI REFERENCES

Main contact: M. Ibrahim | Toronto Metropolitan University | ibrahimshameel89@gmail.com
Affiliations: M. Ibrahim, F. Janabi-Sharifi, and A. Mohammadi - Toronto Metropolitan Univ.; U. Stege and P. Angara - Univ. of Victoria.
Selected DOIs: [6] 10.1109/ACCESS.2022.3157977; [7] 10.48550/arXiv.2508.21361; [8] 10.48550/arXiv.2501.18432.